ECRS 2026 Conference
The 6th International Electronic Conference on Remote Sensing
19–21 October 2026 | Online
MDPI

Towards a Spatio-Temporally Complete Urban Land Surface Temperature Record: A Multi-Sensor Benchmark and Rank-Based Baseline Evaluation for Thermal Downscaling over Barcelona

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INTRODUCTION & AIM

Urban heat-exposure mapping needs high-resolution land surface temperature (LST), yet no single sensor is complete in both space and time:

- **Landsat 8/9:** 30 m grid (100 m native thermal), 16-day revisit, cloud-limited
- **Sentinel-3 SLSTR:** daily, but 1 km
- **ECOSTRESS:** 70 m, irregular revisit
- Standardised, exposure-relevant benchmarks for urban thermal downscaling are lacking.

Aim: build a multi-sensor benchmark and set a verified baseline floor that any learned downscaling model must beat.

METHOD

Benchmark over the Barcelona metropolitan core (464 km², ~21 × 22 km):

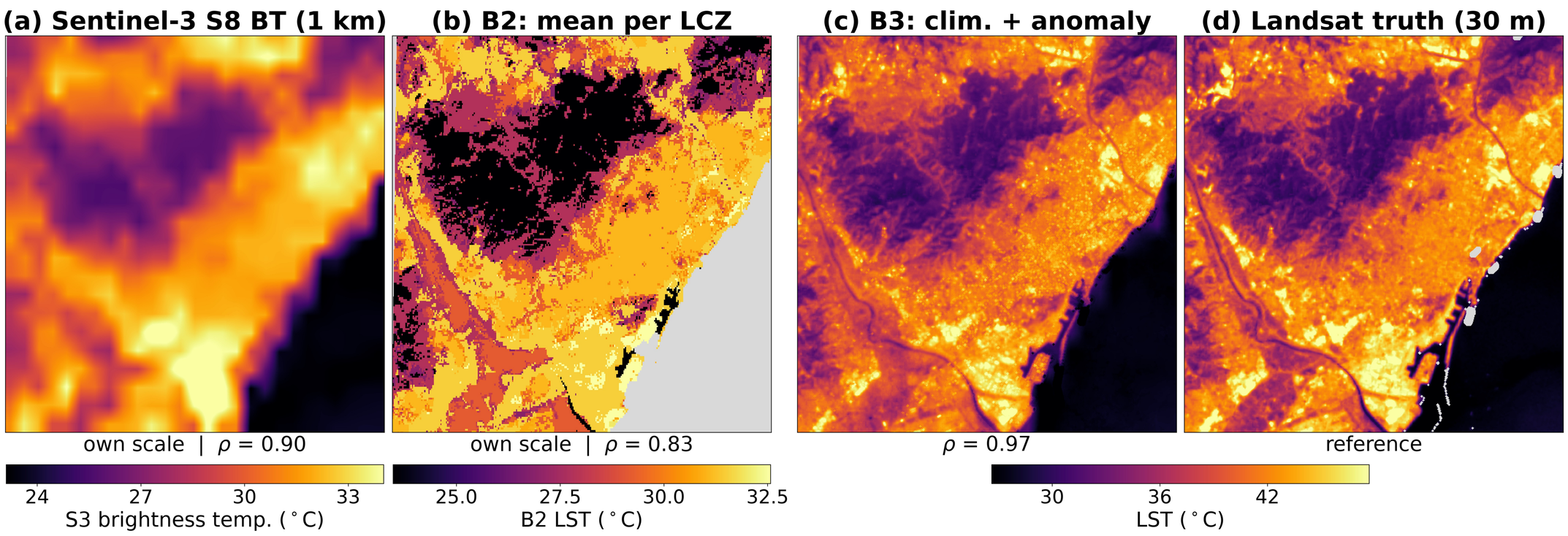
- **189** co-registered pairs, 2016–2024: Sentinel-3 SLSTR S8 brightness temperature (1 km, day pass ≈10 UTC) / Landsat LST (30 m)
- Independent ECOSTRESS (70 m) reference, never used for fitting
- **Temporal split, no leakage:** train 2016–2021 (80) · validation 2022 (34) · held-out test 2023–2024 (75)
- **Rank-based protocol:** Spearman ρ, hottest-10 % IoU, hotspot F1, per-LCZ consistency (exposure depends on spatial ranking, not magnitude)
- **Baselines:** B1 bilinear resample of S3 (no downscaling) · B2 mean LST per Local Climate Zone · B3 = Landsat train climatology + daily S3 anomaly

RESULTS & DISCUSSION

Held-out test, 75 pairs (2023–2024), mean ± s.d. over dates. B1 = S3 without downscaling (reference).

Baseline	Spearman ρ ↑	IoU top-10 % ↑	Hotspot F1 ↑	RMSE (K) ↓	Bias (K)
B1 S3 bilinear resample (no downscaling)	0.73 ± 0.14	0.26 ± 0.13	0.00 ± 0.00	7.10 ± 3.73	−6.42
B2 Mean LST per LCZ	0.62 ± 0.20	0.04 ± 0.02	0.09 ± 0.10	10.22 ± 4.77	−0.20
B3 Landsat climatology + S3 anomaly	**0.79 ± 0.21**	**0.37 ± 0.14**	**0.33 ± 0.17**	**3.98 ± 1.83**	**+1.36**

Chance level for top-10 % IoU ≈ 0.05. B2 does not locate hotspots; B1 ranks well but its top-of-atmosphere values miss every absolute hotspot (F1 = 0).



Held-out date 2024-08-03, per-scene ρ. (a) At 1 km S3 already ranks the scene but spans 11 K vs 21 K, with no 30 m detail. (b) B2 has the LCZ pattern but no daily signal (all-season mean, own scale). (c) B3 recovers the intra-urban detail. (d) Reference.

Cross-check: ECOSTRESS (70 m)

3 test dates with usable swath

Comparison	ρ	RMSE (K)
B3 vs ECOSTRESS	0.79 ± 0.11	4.52 ± 1.23
ECOSTRESS vs Landsat	0.91 ± 0.03	—

n = 3: a consistency check, too small to rank methods.

Why not a learned model (yet)?

- Gradient boosting on raw S3: ρ = 0.78, above the LCZ mean (B2, 0.62), close to B3 (0.79).
- With S3 anomaly + climatology: ρ = 0.81 vs 0.79, not significant (B3 wins 41/75 dates, p = 0.71), with lower RMSE (3.01 vs 3.98 K) and higher F1 (0.43 vs 0.33).
- Static covariates (NDVI, elevation, building density, LCZ) as a residual model on B3: test $R^2 \approx 0$, ρ 0.79 → 0.80. ≈ 97 % of the residual variance is within-pixel, i.e. temporal.

B3 today: a climatology-anchored gap-filler, not a downscaler

OK seasonal ranking of exposure between districts (ρ = 0.79)
OK filling cloudy or non-overpass days
NO day-specific hotspot alerts (F1 = 0.33)
NO absolute daily temperature (RMSE 4.0 K)

CONCLUSIONS

- A high-resolution climatological prior is indispensable: learned models must beat it to add value.
- The B3 residual is temporal: progress needs temporal information, not more static covariates.
- Benchmark, protocol and baselines: a reproducible foundation for spatio-temporal deep learning.

FUTURE WORK/ REFERENCES/ACKNOWLEDGMENT

Future work: evaluate spatio-temporal deep-learning models on this benchmark, against B3.

Data: Sentinel-3 SLSTR, Landsat 8/9 Collection 2, ECOSTRESS L2G LSTE; LCZ map: WUDAPT (Demuzere et al.).

Correction to the abstract: the Sentinel-3 overpass selection was fixed after submission (night → day pass); all results shown use the day pass.

Session S1 — AI in Remote Sensing